Task 1: Motor imagery, what is in the signal and what the networks learn

37 subjects (S088/S092/S100 dropped: different trial timing), runs 4/8/12, 1,662 trials. Cross-subject numbers are means over 37 held-out subjects (5 fixed subject-wise folds), ranges over 3 seeds; all are in `outputs/task1/`.

(a) If a model scores 70% on these subjects, how much could be explained without motor imagery?

Essentially all of it. The cue is a target on the left or right of the screen for 4 s, and subjects look at it.

- **The eyes alone beat the motor cortex.** Within-subject decoders restricted to one route each (A3): frontal channels below 4 Hz reach **71.7%** (25/37 subjects above the $p < 0.05$ chance threshold of 0.644 for 45 trials); sensorimotor mu/beta power reaches **55.3%** (7/37). The posterior visual response to the target adds **60.7%** (13/37). The F7−F8 class difference is an ~80 µV saccade peaking at 0.4 s.
- **The cue sequence leaks too.** Consecutive trials repeat the same class only 25% of the time (random: 50%), so eye position left over from the previous trial predicts the next class before the cue (57.0%, 7/37).
- **The networks follow the eyes.** EEGNet reaches **79.4%** (79.1–79.8) cross-subject, TF-time 77.5%. Per test subject, accuracy tracks the A3 ocular decoder ($\rho = 0.74$, $p = 1e\text{-}7$) and not the A2 lateralisation index ($\rho = 0.07$). With all Part A measures in one regression, the ocular decoder is the only significant predictor ($R^2 = 0.67$). TF-time puts 46% of its attribution on the first 0.9 s, and both models put ~45% of input saliency on 21 frontal sites (1.4× an even spread) vs 14% on the 11 sensorimotor channels (0.8×).
- **What is left without the eyes.** Removing the ocular route (frontal sites dropped, CAR over the rest, 8–30 Hz; checked to take the slow and evoked routes to chance while leaving mu/beta untouched) drops EEGNet to **60.7%** (60.3–61.4) and TF-time to 58.9% (57.3–61.2). Now accuracy follows the A2 µ index ($\rho = 0.55$, $p = 0.0004$): 65.0% in the 18 subjects with lateralised ERD, 56.7% in the 19 without.

So of EEGNet's 29 points above chance, about **19 come from the eyes and the visual response to the target** (the removal closes both, plus any eye-position carry-over from the alternating sequence), and about **11 follow the motor ERD**. A model at 70% on these subjects should be presumed ocular until shown otherwise. Even the residual ~61% is not certified *imagery*: small real hand movements give the same ERD, and there is no EMG. Hemispheric comparisons also depend on the reference, which turns out to sit on the right (the ocular signal is 2–4× larger on left channels under the raw reference, symmetric under CAR); with it, only 12/37 subjects show lateralised ERD vs 18/37 with a surface Laplacian.

(b) Mechanisms behind the B2 results

Scaling (B2.1): flat curves, no crossover. With 5 training subjects EEGNet already reaches 76.1% (96% of its full-data accuracy); TF-time goes 72.6 → 77.5%. A curve this flat means the decisive pattern is the same in every person: a saccade has one spatial field (opposite polarity at F7/F8) and one time course. EEGNet's architecture is that template, a 0.5 s temporal filter followed by a fixed 64-weight spatial filter and a separable conv, so it needs few examples. TF-time has the same front end but must learn from data which of 13 time tokens matter, which costs ~3.5 points at 5–10 subjects and ~2 at 30. A crossover would need structure that varies across subjects beyond what a fixed filter captures; a shared ocular template offers none.

Tokenisation (B2.2). Channel tokens with an identity embedding reach 73.9%; without it **48.2%**, and the training loss stays at ln 2: the model cannot fit even its training data. Without identity, the encoder plus CLS readout is a function of the *unordered set* of channel signals. Left vs right trials differ by a **mirror image**: the saccade is +a at F7 and −a at F8 for one side and the reverse for the other; ERD is low power at C4 for left-hand and at C3 for right-hand imagery. Both give the same multiset of channel signals, so the classes are identical as sets. It matters here: without *where*, no left/right task is solvable, ocular or motor.

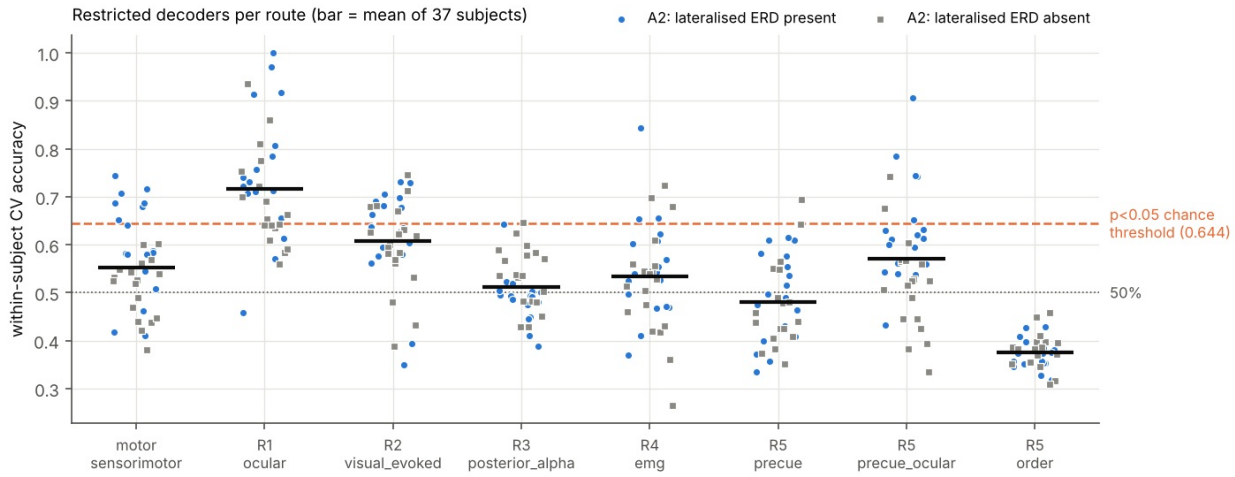
Displacement (B2.3). A 15° lateral cap rotation (24 mm at Cz, ~4 SD of normal placement error) costs only 2.7 (EEGNet), 2.1 (TF-time), 1.1 (TF-chan-id) points; 5° costs ≤ 0.3. It acts mainly as a class bias (TF-time predicts "right" on 55% of trials when the cap slides right, 46% when it slides left). EEGNet loses most because its depthwise spatial filters amplify the input change (relative representation change 0.33 at the input → 0.42 after the spatial conv → 0.45 after the separable conv), while TF-time's 0.3 s patch pooling damps it (0.19 at the tokens, 0.09 at the logits). The ocular topography is broad and changes 14% at 10° vs 29% for the mu topography, but models retrained without the eyes also lose little (−1.4/−0.8 points at 15°; 13%/9% of their margin above chance vs 9%/8%), so this explains only part of the robustness.

Reliance (B2.4). Deleting the 3 most-attended of 13 tokens drops TF-time from 77.6% to 54.1%; the 3 with the highest gradient × input to 68.9%; 3 random ones to 76.0%. *Contrary to my prediction, attention is the more faithful score.* In a two-block encoder read out from the CLS token, the last block's CLS update is literally the attention-weighted sum of the tokens' values, so attention approximates the deletion effect itself. Gradient × input linearly estimates the effect of *zeroing* a token (including its positional encoding), which the softmax renormalisation and LayerNorm make a poor proxy for *removing* it.

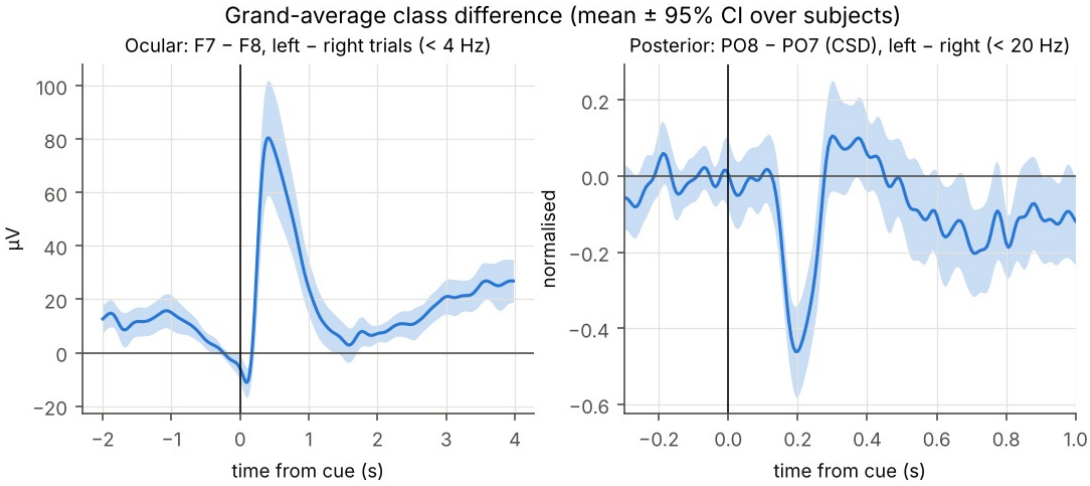
Intervention (B2.6). Without the eyes, the transferable signal is mu/beta power at sensorimotor sites, which imagery changes *relative to each subject's own resting level*, but the networks saw only absolute post-cue power. I divided every channel by its own 8–30 Hz RMS over −1.5…−0.5 s of the same trial, so log power becomes log P(post) − log P(pre) = ERD and any per-electrode gain cancels. Predicted +2–4 points; observed **−1.7** (EEGNet 60.7 → 59.0%) and **−1.6** (TF-time 58.9 → 57.3%), while the correlation with the A2 μ index *rose* (0.55 → 0.59; 0.53 → 0.60). The 1 s baseline has ≈ 44 degrees of freedom, so each channel's normaliser carries ≈ 0.9 dB of independent noise, ≈ 1.3 dB on a C3−C4 contrast whose per-trial signal is ~1 dB; the trial-wide scalar it replaced (43 channels × 4 s) was nearly noise-free. Accordingly, TF-time lost nothing in subjects with lateralised ERD (+0.2) and 3.3 points in those without: the model leans more on ERD but pays for a noisier input. A 4 s baseline would roughly halve that noise.

Predictions that failed (all committed beforehand, `PREDICTIONS.md`): accuracy with 5 subjects (76%, predicted ≈ 60%); the ocular route's strength (72%, predicted ≈ 60%) and the motor route's weakness (55%, predicted 60–62%); a random cue order (it alternates); the size of the displacement effect (≤ 3 points, predicted 5–10); attention less faithful than gradients (the reverse); baseline normalisation improving accuracy (it cost 1.7 points).

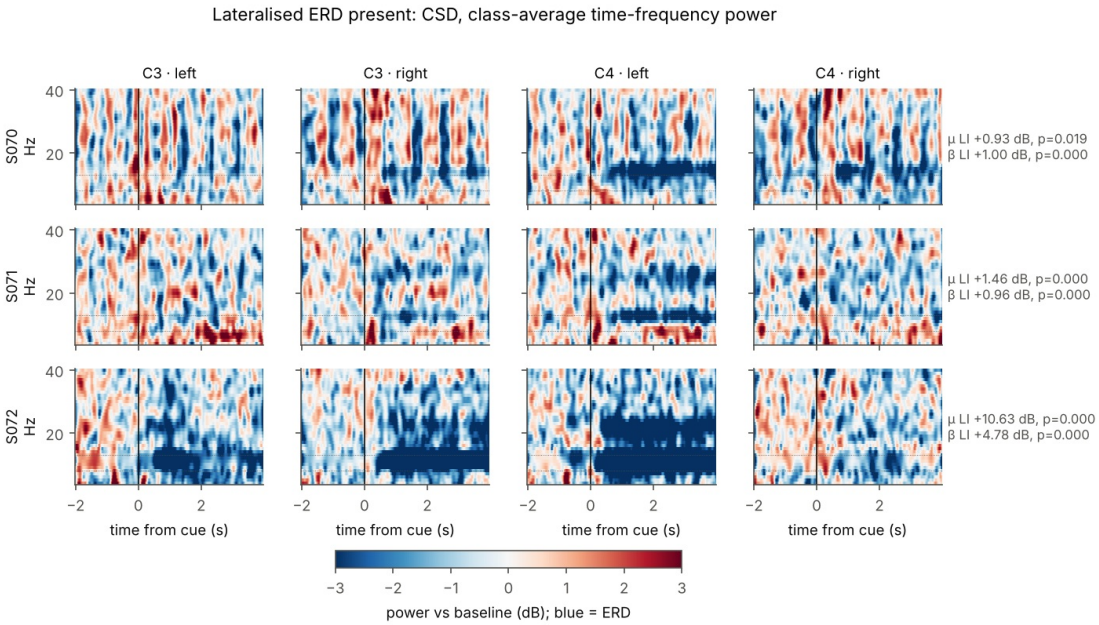
Key figures (all in `outputs/task1/figures/`)



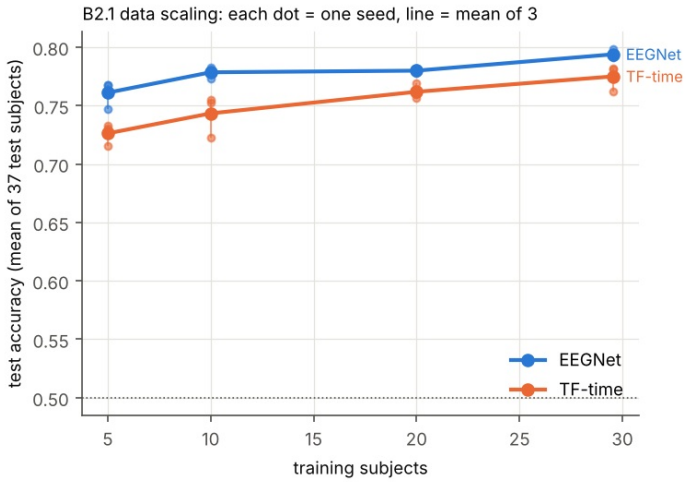
A3. Within-subject decoders restricted to one route each (bar = mean of 37 subjects; dashed = p < 0.05 chance threshold for 45 trials).



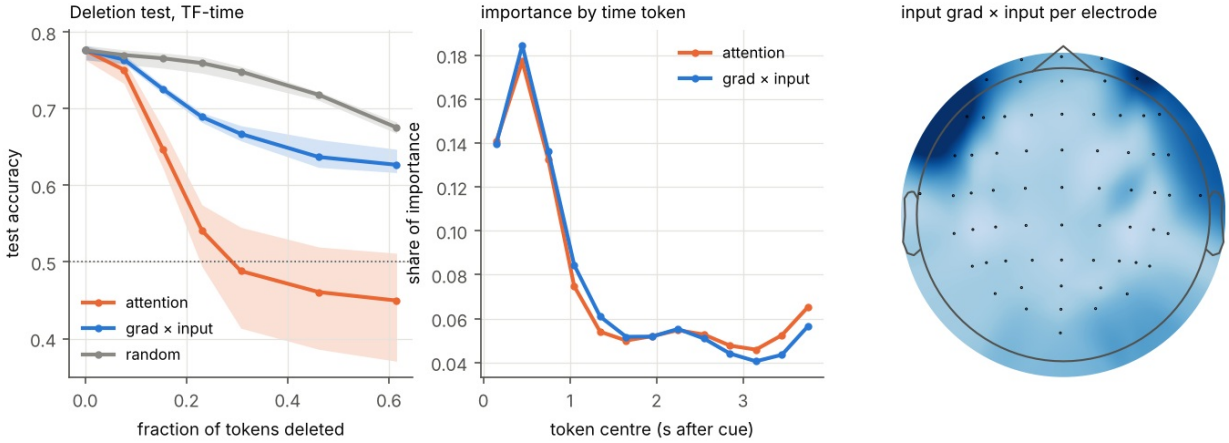
A3. Grand-average left-minus-right class difference: horizontal EOG proxy F7−F8 (left) and posterior PO8−PO7 (right), mean ± 95% CI over subjects.



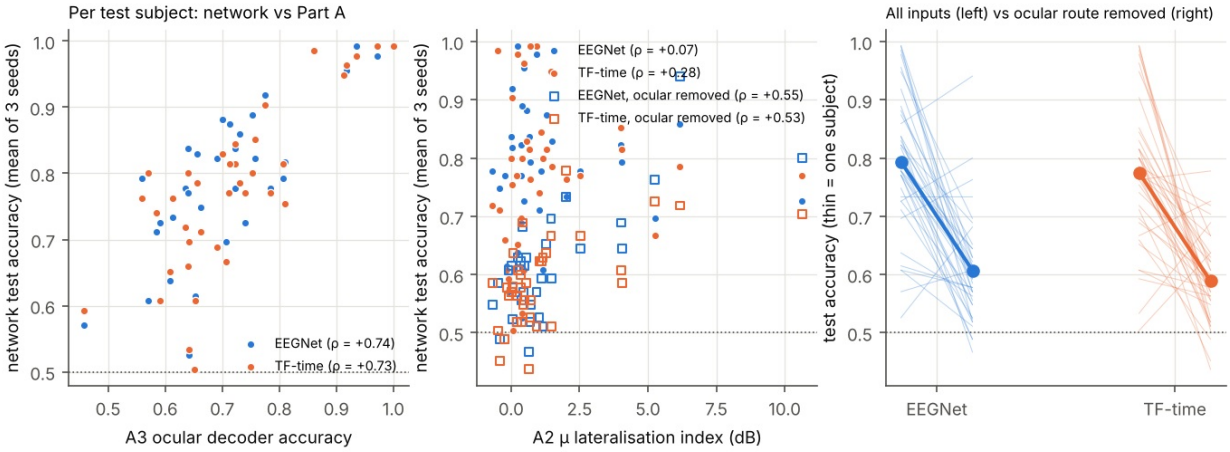
A2. Class-average CSD time–frequency power at C3/C4 for the 3 subjects with the strongest lateralised-ERD evidence (blue = ERD).



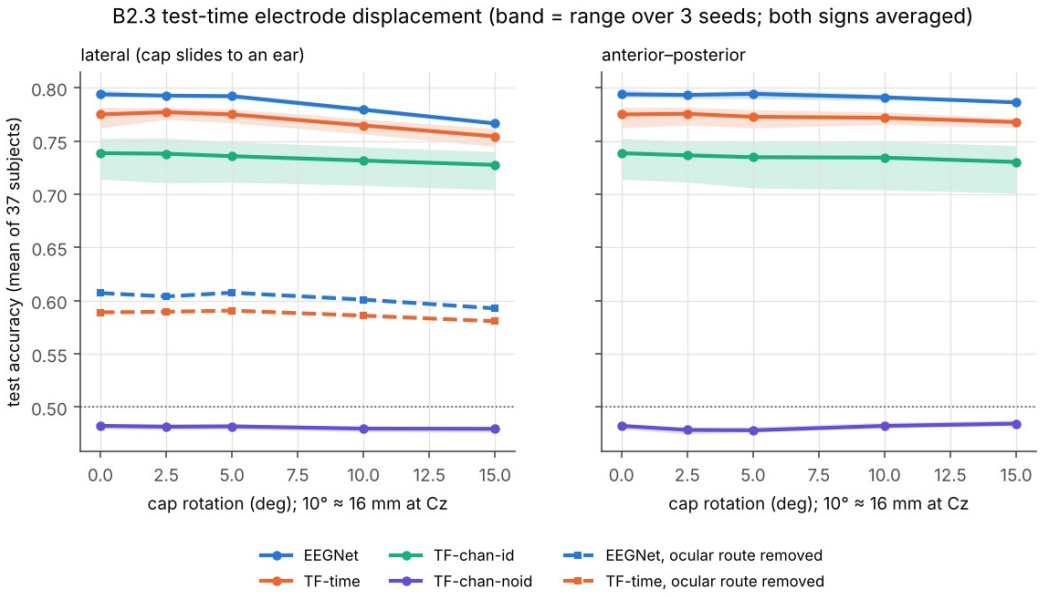
B2.1. Cross-subject accuracy vs number of training subjects; each dot one seed, line = mean of 3.



B2.4. TF-time: deletion test (left), importance per 0.3 s time token (middle), input saliency per electrode (right).



B2.5. Per test subject: network accuracy vs the A3 ocular decoder and the A2 μ index (hollow = retrained without the ocular route); right: all inputs vs ocular route removed.



B2.3. Test-time cap rotation; dashed = models retrained without the ocular route.

Task 2: Sleep staging

Not included in this version of the report. `task2/sleep_pipeline.py` is committed unmodified; the baseline run, defect fixes, impact ledger and the 2d/2e answers are still to be done.